

Computational Thinking Title Here

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Chapter 1

Codes of Life

1.1 Data vs. Information

The dawn of the 21st century fundamentally transformed humanity's relationship with information. What began as simple digitization—converting analog signals into binary—quickly evolved into something far more expansive. Messages no longer required physical delivery. Satellites could monitor the earth's surface in real time. Computing power that once consumed entire buildings now fit in a pocket.

For consumers, this felt like liberation: unprecedented access to information, to each other, to the world. For researchers, it opened new avenues for discovery. Modern data collection and analysis tools have enabled us to gain insights into human behavior, social dynamics, and environmental patterns that were simply impossible a generation ago. These capabilities have revolutionized fields such as epidemiology, urban planning, and climate science.

Yet not all industries embraced this new technology with altruistic intentions. Some companies saw the potential for analyzing human behavior with an implicit understanding: with enough data about a person, you can predict—and perhaps influence—their actions. This line of thinking, perhaps more than any other, drove the rapid rise of data collection. The practice became so pervasive that it often goes unnoticed by the individuals whose data is being gathered.

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Yet, not all industries embraced this new technology with the same altruistic intentions. Some companies saw the potential for analyzing human behavior with the implicit understanding that with enough data about a person, they could predict and perhaps influence their actions. This line of thought, perhaps more so than any other has led to the rapid rise of data collection. This

collection has become so pervasive that it often goes unnoticed by the individuals whose data is being gathered.

Mobile phones and devices like them have become not simply tools for communication or convenience, but via applications, become instruments of observation. Every search query logged. Every location ping recorded. Every purchase, every pause, every click—collected, stored, and increasingly, sold. The question worth sitting with is not *how* this happened, but *when we agreed to it*—and whether we ever truly did.

As sensors proliferated and storage costs collapsed, we crossed a threshold into the era of *big data*—defined not just by volume, but by the unprecedented velocity at which information arrives and the variety of sources from which it flows. Social media posts, smartphone location traces, financial transactions, scientific instruments, surveillance cameras: each generates continuous streams of data points accumulating at scales previously unimaginable. The grand ambitions emerging from Silicon Valley cast data as the new frontier of economic and social transformation—the proverbial gold rush of the digital age.

The looming question is **why?**. What makes data so valuable? At what moment did we decide that data about human behavior was more valuable than the devices generating it? How did we reach a point where a mobile phone, worth several hundred dollars, was worth so much less than the data it generated? Why drives all these companies and organizations to collect and monetize data? Importantly, how do we go from the idea of collecting data to the point where it can become an actionable tool for prediction?

In response to these very questions, entire fields have emerged to help us understand and interpret the data we encounter and choose to collect. Disciplines such as statistics, computational thinking, data science, and machine learning offer tools and frameworks for extracting meaning from vast datasets. But before any of that becomes useful, we need to ask a more fundamental question—one that turns out to be surprisingly difficult to answer cleanly:

What actually is data? And when does it become meaningful?

To answer these questions meaningfully, we must first understand what “meaningful” actually requires. Data does not arrive pre-interpreted. It must be structured, contextualized, and analyzed before it yields insight—whether through descriptive statistics, predictive modeling, or simply careful reading. The gap between raw data and actionable understanding is precisely what this textbook is concerned with.

Consider the most immediate data available to you right now: the text on this page. It may seem like an odd example, but ask yourself genuinely: *How do you interpret the symbols before you? What makes this text meaningful rather than noise?*

The intuitive answer is language—but that answer conceals a great deal. Consider the following sequence of symbols:

THEOSTRICHCANNOTFLY

You likely parsed that relatively quickly: *The ostrich cannot fly*. But nothing in the string

itself told you where one word ends and another begins—no spaces, no punctuation, no explicit boundaries of any kind. You supplied the rules yourself: word boundaries, grammatical structure, semantic plausibility, drawing on a lifetime of learned context applied so automatically you may not have noticed doing it.

Now consider the same symbols, differently arranged:

THEO STRICH CAN NOT FLY

Suddenly: *Theo Strich cannot fly*. The data is identical. A single structural rule—where spaces appear—has changed the meaning entirely.

The production of data is easy; the challenge lies in interpreting and deriving **meaning** from it.

Consider the example of three civilizations, each with its own varying complexities in how data was transcribed.

The first would be humanity’s humble beginnings, where early humans used simple symbols and drawings to communicate and record their experiences. This is largely considered a key marker of when humans began to think in abstract terms—a precursor to the development of complex language and writing systems.

These early forms of data were carefully drawn symbols etched into caves, walls, or stone. A 2026 finding on the Indonesian island of Sulawesi dates back nearly 67,800 years. These handprints and animal silhouettes begat questions humanity fundamentally will likely never answer: *why*. We see it and assume it was a signature, a marker of presence—“I was here.” But was it? Was it a ritual? A warning? A record of a hunt? A spiritual practice?

This is the fundamental fragility of data: without context, it becomes a guessing game. The cave painter had perfect information—they knew exactly what they meant, if they meant anything at all. But we, the interpreters separated by millennia, possess only the data. We can analyze pigment composition, dating techniques, spatial arrangement, yet perhaps the most important question of all remains trapped in silence.

Such systems naturally evolve with the needs and capabilities of the societies that create them.

We see this evolution in the transition from cave paintings to hieroglyphs invented by the ancient Egyptians.

Unlike the ambiguous handprints of Sulawesi, hieroglyphs were *designed*. They arose from a deliberate need: to record and communicate administrative and religious information with standardization and clarity. Here was humanity’s first attempt to solve the context problem—to create a system where the same symbol would mean the same thing to everyone who encountered it.

This was done through *organization* and *structure*. Consider two identical sets of hieroglyphs arranged differently as shown in Table 1.1.

An individual hieroglyph carved in isolation tells us almost nothing. But a collection of hieroglyphs arranged in sequence, organized by rules, structured in rows and columns—that structure

Table 1.1: Identical hieroglyphs gain different meanings through structural context

Context/Structure	Same Glyph	Resulting Meaning
Isolated on temple wall	 (Eye of Horus)	Ritual symbol, divine protection
In chronological sequence	 (position 3 of 5)	“Third event: the watching”
In tax ledger column	 (quantity: 12)	“12 units of temple offering”

creates meaning. The data is identical. The structure transforms it into different information depending on how it’s collected and organized. What we choose to measure, when we measure it, and under what conditions—along with how we organize and encode relationships within the data—fundamentally determine its meaning and usefulness.

The key understanding is that data is not inherently meaningful; it requires context, organization, and structure to become **information**.

Definition — Data vs. Information

- **Data:** Raw symbols, measurements, or observations without inherent meaning
- **Information:** Data + structure + context + interpretation

This is the critical insight: **information—not raw data—is what ultimately drives decisions**.

Raw data sitting in a warehouse, uncollected with care or organized without intention, is inert—incapable of informing decisions on its own.

Information—data that has been thoughtfully collected, properly structured, and interpreted within context—is what actually moves decisions, drives strategy, and creates value.

The difference between raw data and actionable information is often the difference between failure and success. This textbook will impart the principles and practices of data management, collection, analysis, visualization, and interpretation. In doing so, it will also challenge readers to ask fundamental questions about the role of data in our lives and the world around us. Without this understanding, we cannot fully appreciate the power and potential of data in shaping our world. Without it, we risk causing incredible harm by misinterpreting, incorrectly collecting, or misusing data.

Each chapter will explore real-world applications of these concepts through the three civilizations we have examined. Astute readers might notice that we have yet to formally introduce our third civilization, but it should come at no suprise that our final civilization—the one navigating the data deluge of the 21st century—is *our own*.

1.2 Asking Questions About Data

In Section 1, we established that data becomes information only through structure, context, and interpretation. But interpretation does not happen automatically—it begins with a question.

Consider the symbols you just read: letters formed into words, words into sentences. The data was constant; the meaning emerged when you asked something of it: *What does this say? What is the author trying to tell me?*

Asking questions of data is no different. The data sits silent until we interrogate it. Yet the quality of our answers depends entirely on the quality of our questions.

Artificial intelligence has transformed the landscape of data analysis. Models can now process datasets of unimaginable scale, identify patterns invisible to the human eye, and generate insights in seconds that once took teams of analysts weeks to uncover. Yet these capabilities are only possible due to a detailed understanding of our data and the purpose behind the questions we choose to ask. Recent studies (2025) note that models struggle with strategic planning quality. They excel at single instructions, but as of today, they fail to independently chain together the logical steps required for complex inquiry.

This limitation reveals something deeper: asking good questions is hard precisely because meaningful questions are inherently ambiguous. Consider a question like “Why is churn increasing?”—a human analyst recognizes the ambiguity and responds by asking clarifying questions or drawing on domain context to narrow the scope. A model, lacking this contextual understanding, cannot distinguish a meaningful question from a poorly formed one. The result: it confidently pursues meaningless correlations because it does not “understand” the business context or the ambiguity embedded in the request.

This is why the human ability to inquire—to recognize ambiguity, to ask clarifying questions, to draw on experience and context—remains irreplaceable. AI excels at *answering* questions. But it cannot tell you whether you asked the *right* question.

To get at the best questions, we must start with the highest order question: *What is the unknown?*. This can be phrased however you might like, the core idea is to stimulate curiosity and encourage deeper thinking about the data at hand. Asking questions and solving problems are practical skills that require practice and experience. Much like any practical skill, writing, reading, sports, we acquire these abilities through consistent practice and real-world application.

Before diving into the specifics of statistical analysis, let’s first explore four fundamental steps in asking effective questions outlined in G. Polya’s book “How to Solve It”.

Key Terms

These terms will anchor our discussion:

- **Hypothesis:** The underlying claim or relationship we are trying to understand or prove.
- **Decomposition:** Breaking a complex question into smaller, answerable parts.

1.2.1 Understanding The Problem

It is tempting to believe that all problems can be solved with data. This sentiment is understandable—even forgivable—given the breathless enthusiasm surrounding big data’s potential. Silicon Valley has, at times, promoted data as a kind of universal solvent: apply enough of it to any problem and answers will emerge. This line of thought is a dangerous one, without understanding the problem and the wider context, data can and will be misused or misinterpreted. This often arises from hubris directly as a result of data scientists and engineers who overestimate their own understanding of the problem and the data at hand.

Real World

In 2008, Google launched *Google Flu Trends*, a system designed to predict influenza outbreaks by analyzing search query patterns. The premise was elegant: if people are getting sick, they will search for flu symptoms, and those searches would appear in the data before hospital admissions did. For several years, the system appeared to work. Then, during the 2012–2013 flu season, it overestimated infections by nearly double. The root cause was not bad data—it was a bad question. The team had asked “*Does search volume predict flu?*” when the more fundamental question—“*Does search volume measure illness, or awareness of illness?*”—had never been posed. These are not the same thing, and no model can tell the difference if the question is never asked.

This failure had a name. Researchers writing in *Science* called it “big data hubris”: the assumption that data volume alone substitutes for careful problem understanding. It remains one of the most instructive cautionary tales in the field precisely because the data was abundant, the engineering was sophisticated, and the question still came out wrong.

Polya’s first step is deceptively simple: *understand the problem*. In practice, this means resisting the impulse to reach for data immediately. Before selecting a model, writing a query, or cleaning a dataset, the practitioner must first ask: *What is the unknown? What are we actually trying to learn?* Only then can we ask a harder follow-up question—one that turns out to matter enormously in practice: *Can this question even be answered with data?*

Yet understanding the problem is a challenging task. How does one build institutional knowledge about something unknown? Is that not what we are trying to do when we ask questions about data? To do this, we can break our phase of understanding into several key components:

- Isolate the principal parts of the problem.
- Identify the core issue or phenomenon.
- Evaluate the feasibility of answering these questions with available data.
- Gather background information and context.

Checkpoint

Pause and reflect: What problem are we *really* trying to solve when we ask questions about data? In the Flu Trends case, Google asked “Can search volume predict flu?” but should have asked “Does search volume measure actual illness?” What unknown are *you* chasing in your own analysis?

1.2.2 Isolating the Principal Parts of a Problem

The ability to isolate the principal parts of a problem is similar to outlining an essay. Before writing, you identify your main points and organize them logically. Each point builds on the last, transitioning smoothly from one idea to the next. The end result: a coherent argument that communicates your original intent.

Asking questions of data works the same way. It’s a scientific approach to breaking down a complex hypothesis into smaller, manageable components that build toward comprehensive understanding.

Consider a simple essay prompt: “*How did the printing press change European society?*”

A well-written essay doesn’t answer this directly. The writer would break it down:

- What was information dissemination like before the printing press?
- How quickly did printing technology spread across Europe?
- Which social groups gained access to written materials?
- What new ideas emerged that couldn’t have spread before?

Each sub-question is narrower. Each is answerable with specific evidence. Together, they build toward answering the original prompt.

The hypothesis is ultimately the principal reason we might be interested in some dataset. It’s the underlying thing we are trying to understand or prove.

Consider Google Flu Trends again. The principal hypothesis was that search volume could predict flu infections. The unknown relationship: the causal link between search behavior and actual illness.

But this single question hides multiple unknowns:

- Do people search for flu symptoms when they get sick?
- Do search patterns change during flu season, or only during outbreaks?
- Are searches correlated with actual infections, or just awareness of flu?

The Google team asked the first question. They never asked the third. That omission cost them dearly.

Isolating principal parts forces you to clarify what you're actually trying to prove—and to determine whether your data can prove it. Datasets rarely yield challenging answers easily. The Flu Trends analysis required layers of examination: the initial hypothesis broken into components, each component tested through various combinations, the results analyzed and re-analyzed.

This is the foundation of scientific inquiry. You move from curiosity about a phenomenon to a carefully framed question that data can address. Without this understanding, even sophisticated tools produce misleading conclusions.

The hardest part of understanding a problem is finding the right *first* question to ask. That requires deep consideration of the problem's principal parts.

Checkpoint

Try it yourself: Take a complex question you're curious about. Break it into 3–4 smaller, specific parts. Which parts are directly answerable with data? Which require inference or judgment?

1.2.3 What Makes a Question Answerable?

Not every meaningful question is a data question. To see why, consider one of the earliest systems for near-real-time market data: the ticker tape.

Invented by Edward A. Calahan in 1867, the ticker tape machine used telegraph infrastructure to print stock symbols, prices, and trade volume onto a narrow paper strip—effectively a one-way market data feed from the exchange floor to brokerages across the country. By 1929, ticker machines were widespread and had materially improved market information flow. But they carried a hard physical constraint: even improved models could relay data at roughly 285 characters per minute. When trade volume surged beyond that throughput, the tape fell behind.

During the crash of October 1929, that lag grew from minutes to hours—roughly 100 minutes on October 21, and nearly four hours by October 24. This delay ultimately proved catastrophic.

Picture a trader in a brokerage office on Black Tuesday. Their ticker tape clicks out a price: \$30 per share for U.S. Steel. The trader makes their move, selling what they think is \$30 worth of stock. But that tape is four hours behind. On the exchange floor, panic selling has already crushed the price to \$10. The trader didn't sell at \$30. They sold at \$10 and lost three-quarters of their money without knowing it until the next day's paper confirmed the disaster.

The ticker tape showed \$30. The floor price was \$10. The data wasn't wrong—it was just old. And in a market crash, old data might as well be fiction.

Panicked traders kept asking questions like “What is the current price?” and “How much has it fallen?” But the system couldn't answer those questions. It could only tell them what the price *had been*—hours ago. Investors sold based on outdated information, making decisions that felt rational but were grounded in data that no longer described reality.

Traders were making decisions on prices that no longer existed. The system never stopped

producing data. It simply stopped producing *timely* data, and timeliness was the very context that made the data meaningful.

Now consider what a data analyst in 1929 could and could not have answered:

- “Did ticker tape reporting delays exceed 60 minutes during October 1929?”
- “Was the 1929 stock market crash caused by ticker tape delays?”

The first question was a directly measurable one at the time. An analyst would have been able to answer the first question by examining when a signal was sent from the exchange floor and when it was received by the ticker tape machine. This would provide a clear measure of the delay.

The second question, however, is far more complex. Based on the narrative above, it *seems* likely that ticker tape delays were the primary cause of the crash. However, this is a causal question that goes beyond what the data can directly tell us. Ticker tape data alone offers no insight into the economic conditions, human psychology, regulatory failures, and cascading feedback loops that no dataset fully captures.

Asking that question of the data will not produce a wrong answer so much as a dangerously incomplete one.

A question is answerable by data when it satisfies three conditions:

1. **Observability:** The phenomenon can be measured or recorded in some form.
2. **Specificity:** The question defines clear conditions for what counts as an answer.
3. **Falsifiability:** There exists some possible observation that could prove the answer wrong.

Questions that fail these criteria are not “bad” questions—they are simply not data questions. “Why did investors panic in 1929?” is a profound historical question, but no amount of trading volume data can fully answer it. What data can do is show us *when* panic began, *how fast* it spread, and *which* securities were affected first.

Key Terms

Now that we’ve seen what makes a question answerable, here are three key criteria:

- **Observability:** Whether a phenomenon can be measured or recorded.
- **Specificity:** Whether a question defines clear conditions for what counts as an answer.
- **Falsifiability:** Whether some possible observation could prove the answer wrong.

Key Insight

Data answers “what” and “when” questions reliably. It answers “why” questions only through careful inference and modeling—and even then, with uncertainty.

Checkpoint

Synthesis: Both Google Flu Trends and the 1929 ticker tape failed for the same reason: the data existed, but it didn't measure what people *thought* it measured. Flu searches didn't equal flu cases. Ticker prices didn't equal current market value. What's the gap between what your data *does* measure and what you *need* it to measure?

1.2.4 The Role of Context

Recall from Section 1 the identical hieroglyphs that meant different things depending on their structure. The importance of understanding the context of how a dataset was collected and what bias it might contain cannot be overstated. Many of the most high-profile examples of data and algorithmic misuse have been due to this very reason. As they say: “The road to hell is paved with good intentions.”

Consider an algorithm designed to predict the likelihood of breast cancer from medical imaging. The hypothesis is straightforward: if trained on enough historical examples, a model could learn to identify the visual markers of malignancy and flag high-risk cases earlier than a human radiologist might catch them. The unknown is the degree to which the algorithm can generalize—whether patterns learned from past cases transfer reliably to new patients.

To test this hypothesis, we need a large labeled dataset: historical scans marked as malignant or benign, drawn from clinical records and published studies. The more data, the better. Or so the thinking goes.

Checkpoint

Before reading further, consider the dataset described above— historical medical records and clinical studies.

1. Who has historically had access to high-quality clinical imaging and follow-up care? Who has not?
2. If certain populations are underrepresented in the training data, what happens when the algorithm encounters them?
3. The goal is a generalizable model. What assumptions are embedded in using historical records to build it?

Real World**Mammography AI and Racial Disparity**

In 2024, researchers published a study in *Radiology* examining a commercially available, FDA-approved AI algorithm used to assist radiologists in reading screening mammograms. They analyzed more than 4,800 mammograms that had already been confirmed as cancer-free by physicians—meaning any case the algorithm flagged as suspicious was, by definition, a false positive.

The results were striking. Black women were 50 percent more likely to receive a false positive result than white women. The algorithm repeatedly flagged tissue as suspicious in patients who did not have cancer—at significantly higher rates for Black patients than for any other demographic group in the study.

The cause was not a flaw in the algorithm’s code. It was a flaw in the data it learned from. AI breast cancer tools are predominantly trained on imaging datasets drawn from majority-white patient populations. When those models encounter mammograms from Black women—who on average have denser breast tissue and different tissue characteristics—the algorithm has less experience with those patterns and performs less reliably.

The consequences are real. A false positive triggers a recall: additional imaging, possible biopsy, anxiety, and financial cost. As the study authors noted, if radiologists normalize AI recommendations calibrated primarily on white patients, they risk driving higher recall rates for Black patients—worsening the very health care disparities the technology was meant to reduce.

The FDA, at the time of the study, had no requirement that AI software be tested on a demographically diverse population before approval.

The altruistic intent behind the development of AI in healthcare is commendable. Now, and more so in the future, these systems will provide significant benefits to patients and healthcare providers. This idealized outcome is only achievable if we carefully consider the context and potential biases in the data. This understanding, however, is **not** always easy to achieve. Again, this often happens due to a disconnect between the subject matter experts and a lack of understanding about the problem set itself. Importantly, the accuracy of the tested model depends entirely on the quality and representativeness of the training data. The algorithm developers likely did not have a full understanding of the data’s limitations or the potential for bias. The data itself was correctly collected, but it was not representative of the diverse patient population that would ultimately use the AI system. The data didn’t lie. But it didn’t tell the whole truth either.

Understanding context means understanding the gaps—what’s missing, whose voices aren’t represented, what got smoothed over in the name of cleanliness.

Context is not simply understanding the data itself, but the world in which the data lives and the ultimate results it produces. As an example, consider the question: “Is 50 a high value?” Without context, the question is unanswerable. Fifty what? High compared to what? High for whom, and for what purpose?

- 50 shares traded on the NYSE in 1929: *vanishingly small*
- 50 shares traded by a retail investor today: *typical*
- 50 million shares traded in a single block: *institutional scale*

The number is identical. The context transforms its meaning.

Context determines:

- **Relevance:** Does this data pertain to the question at hand?
- **Scale:** What magnitude are we working with, and what is normal?
- **Timing:** When was this data collected, and does it still apply?
- **Purpose:** Who collected this data, and what decisions might it inform?

The 1929 crash illustrates the cost of missing context. Investors reading delayed ticker tape saw falling prices but lacked the context of *why* prices were falling, *how many* others were selling, and *what* the actual state of the market was. They acted on incomplete information—not because the data was wrong, but because the context was missing.

While this book will try to emphasize the importance of context in data analysis, to truly grasp the implications of this, I encourage readers to pick up a copy of *Weapons of Math Destruction* by Cathy O’Neil.

Key principle: A question without context is a map without a scale. You may know where you want to go, but you cannot know how far it is, or whether you have arrived.

1.2.5 The Question Hierarchy

Effective data inquiry follows a natural progression. Each level builds on the one below it:

1. **Descriptive:** “What happened?” (e.g., “What was the trading volume on Black Tuesday?”)
2. **Diagnostic:** “Why did it happen?” (e.g., “Why did volume spike on October 29?”)
3. **Predictive:** “What will happen?” (e.g., “Will high volume persist tomorrow?”)
4. **Prescriptive:** “What should we do?” (e.g., “Should we sell, hold, or buy?”)
5. **Reflective:** “Should we have done something differently?” (e.g., “Was our risk model adequate?”)

Each level requires more context, more assumptions, and more judgment. Descriptive questions are the most answerable with data alone. Prescriptive and reflective questions require human reasoning layered on top of data-driven insights.

AI excels at levels 1–3. Levels 4–5 remain firmly in the human domain—because they require not just answers, but *wisdom* about what answers mean and what actions they justify.

Bibliography